The Effect of Pruning on Decision Tree Performance

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Abstract

Decision trees are popular machine learning models due to their interpretability and ability to handle both categorical and numerical data. However, they often overfit, leading to poor generalization. In this project, we evaluate the performance of decision trees for classification and regression tasks on six datasets, and apply reduced-error pruning to reduce overfitting. We compare these models against a Null Model and assess performance with 5×2 cross-validation. Our findings confirm that decision trees consistently outperform the Null Model, with reduced-error pruning often yielding further performance gains while substantially reducing model complexity. We also observe a positive correlation between tree size reduction and performance gains within classification and regression tasks.

Keywords: Decision trees, reduced-error pruning, ID3, C4.5

1 Introduction

As machine learning models become more complex, one issue they often run into is a lack of interpretability. Decision trees strike a rare balance in this regard: they are easy to interpret and visualize, but are also surprisingly effective on many tasks. In particular, decision trees are essential in fields such as credit risk assessment and medical diagnoses, where understanding the decision-making process is crucial.

But decision trees are also vulnerable to overfitting. One way to mitigate this is by introducing *reduced-error pruning*, wherein branches that do not contribute to performance on a pruning dataset are iteratively removed. While pruning attempts to make the tree more generalizable to unseen data, it may also discard useful patterns. Therefore, there is a tradeoff to be explored.

In this project, we use 5×2 cross-validation to evaluate the performance of decision trees, with and without pruning, on a variety of classification and regression tasks. We compare the performance of our trees with that of a *Null Model* that always predicts the majority class (for classification asks) or the mean target value (for regression tasks). We test the following hypotheses:

1. Unpruned decision tree models will significantly outperform the null model across all classification and regression tasks.

2. Pruned decision trees will significantly outperform unpruned decision trees, due to better generalization.

3. There will be a positive correlation between the percentage reduction in tree size and the percentage increase in performance for pruned vs. unpruned trees.

2 Overview of Datasets

The main characteristics of each dataset are listed below.

Table 1: Summary of all datasets

Dataset	**Size (rows)**	**# Features**	**Task**
Breast Cancer	699	9	Classification
Car Evaluation	1728	6	Classification
Congressional Voting Records	435	16	Classification
Abalone	4177	8	Regression
Machine CPU	209	6	Regression
Forest Fires	517	12	Regression

3 Pre-Processing

Each dataset was imported and subjected to the following pre-processing steps:

- **Missing Values**: Missing values were identified using dataset-specific markers and imputed based on feature type. Numerical features were imputed using the mean of the feature. There were no missing values for categorical features.
- **Column Filtering**: Unimportant columns such as unique identifiers and regression prediction columns that could interfere with our predictions were discarded.
- **Cyclic Encoding**: In *Forest Fires*, the "day" and "month" features represent points on a cycle rather than linear or unordered categories. Encoding them as categorical variables would ignore this structure, while assigning them integer values would impose a false notion of linear distance. To reflect their cyclical nature, we encoded each as a pair of sine and cosine values, enabling the model to capture seasonal patterns more effectively.
- **Transforms**: In *Forest Fires*, the target ("area") feature was heavily skewed towards 0. Following the dataset authors' recommendation, we applied a log transformation of the form $\hat{\mathbf{r}}(x) = \ln(\mathbf{r}(x) + 1)$ where $\mathbf{r}(x)$ is the area associated with example x. All regression predictions and evaluations were performed in this transformed space.

4 Algorithms

To construct decision trees from each dataset, we built an algorithm based on the C4.5 algorithm, which extends the classic ID3 algorithm to handle both categorical and numerical features. The algorithm makes use of a metric known as *information gain ratio* which discourages overfitting by penalizing high-arity splits. The information gain ratio is defined as the information gain divided by the intrinsic value of a feature, as defined below:

$$gain(f_i) = \left(-\sum_{\ell=1}^{k} \frac{c_{\pi,\ell}}{|\mathcal{D}_\pi|} \log \frac{c_{\pi,\ell}}{|\mathcal{D}_\pi|} \right) - \left(\sum_{j=1}^{m_i} \frac{|\mathcal{D}_\pi^j|}{|\mathcal{D}_\pi|} \mathcal{H}(\mathcal{D}_\pi^j) \right)$$

$$IV(f_i) = -\sum_{j=1}^{m_i} \frac{|\mathcal{D}_\pi^j|}{|\mathcal{D}_\pi|} \log \left| \frac{|\mathcal{D}_\pi^j|}{|\mathcal{D}_\pi|} \right|$$

In the equations above, $\mathcal{D}_\pi$ is the subset of data in partition π, $c_{\pi,\ell}$ denotes the number of examples in partition π that occur in class ℓ, $\mathcal{H}(\mathcal{D}_\pi)$ is the entropy of partition $\mathcal{D}_\pi$, and m_i is the number of domain values in feature f_i.

The algorithm works as follows:

```
Initialize the root node with the training dataset.

At each node, evaluate each feature to pick a splitting feature:
- If classification:
  - If categorical feature, consider one branch for each feature value.
  - If numerical feature, sort the class labels and identify candidate.
    thresholds at midpoints where the class label changes.
  - Choose the feature and threshold maximizing information gain ratio.
- If regression:
  - If categorical feature, consider one branch for each feature value.
  - If numerical feature, sort class labels into quartiles and choose
    the midpoints between quartiles as candidate thresholds. If there
    are less than 4 samples, all samples generate a new branch.
  - Choose the feature and threshold minimizing total squared error.
- Split the node using the best feature and threshold.

Repeat the process recursively on each child until:
- All data points at the node belong to the same class (classification)
  or the same target value (regression).
- All proposed splits have a zero gain ratio (for classification).
```

To prune our decision trees, we use *reduced-error pruning*, which requires a separate pruning dataset, and works as follows:

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Evaluate the unpruned tree on the pruning dataset.

Repeat the following until no further improvement is possible:
- Identify all non-leaf nodes as pruning candidates.
- For each candidate:
    - Temporarily prune the node by replacing its subtree with a leaf.
    - Assign the new leaf a prediction. (majority class or mean value).
    - Evaluate validation performance:
        - For classification: compute accuracy.
        - For regression: compute mean squared error.
    - Restore the original subtree.
- Permanently prune the candidate with the best performance gain.
```

Note: To improve efficiency, during the first evaluation, the path that each sample from the pruning set takes in the decision tree is cached. Then, when evaluating candidate pruned trees, only samples affected by a pruning candidate are recomputed, and the performance of these samples is merged with the cached values for the rest of the tree. This avoids re-evaluating the entire tree after each tentative prune.

5 Experimental Approach

For each dataset, we performed 5×2 Cross-Validation to evaluate the Null model, the Unpruned Decision Tree model, and the Pruned Decision Tree model. Although maximum tree depth and other stopping criteria are common hyperparameters in real-world decision tree training, we followed the project specification and grew trees to full depth, using reduced-error pruning as the sole form of complexity control. Therefore, there were no hyperparameters to tune. The cross-validation approach is outlined below:

- Import the dataset and apply the pre-processing steps outlined in Section 3.
- Divide the dataset into 80% - 20% splits, for training/testing and pruning, respectively.
  - For classification datasets, we stratify by class label to ensure approximately equally distributed classes in each split.
  - For regression tasks, we sample uniformly across the target values.
- Repeat the following five times:
  - Divide the training/testing set into two equally sized partitions (stratified or uniformly sampled).
  - For each partition:
    - Build a decision tree using this partition as the training data.
    - Apply regression/classification on the other partition using this tree, and store the performance results.

  - Apply reduced-error pruning to the decision tree.
  - Apply regression/classification on the other partition using the pruned tree and store the performance results.
  - Apply the Null Model on the other partition and store the performance results.

- Average the results of each experiment for each candidate model.

Note: When evaluating performance for classification, we use the *Error Rate* (percentage of predictions that are incorrect), and for regression we use the *Mean Squared Error*.

6 Results

Table 2: Average Error Rate across all models for classification tasks.

Average Error Rate			
Dataset	**Null Model**	**Unpruned Tree**	**Pruned Tree**
Breast Cancer	0.3448	0.0618	0.0495
Car	0.2998	0.0919	0.1010
Congress	0.3862	0.0616	0.0506

Table 3: Average Mean Squared Error (MSE) across all models for regression tasks.

Average MSE			
Dataset	**Null Model**	**Unpruned Tree**	**Pruned Tree**
Abalone	10.3927	9.1850	6.5788
Machine	25663.9987	7143.1009	6095.6850
Forest Fires	1.9518	3.7625	2.5902

Table 4: Wilcoxon Signed-Rank Test p-values for classification datasets.

Dataset	**Unpruned vs. Null**	**Pruned vs. Null**	**Pruned vs. Unpruned**
Breast Cancer	0.001	0.001	0.039
Car	0.001	0.001	0.999
Congress	0.001	0.001	0.008

Table 5: Wilcoxon Signed-Rank Test p-values for regression datasets.

Dataset	**Unpruned vs. Null**	**Pruned vs. Null**	**Pruned vs. Unpruned**
Abalone	0.001	0.001	0.001
Machine	0.001	0.001	0.042
Forest Fires	1.000	1.000	0.001

Table 6: Tree Size Comparison Before and After Pruning

Dataset	**Average # Nodes (Unpruned)**	**Average # Nodes (Pruned)**	**% Reduction in Tree Size**
Breast Cancer	42.8	20.4	52.3%
Car	241.2	150.9	37.4%
Congress	32.8	14.5	55.8%
Abalone	2953	1234.4	58.2%
Machine	191.2	95	50.3%
Forest Fires	357	145.4	59.3%

7 Discussion

Tables 2 and 3 summarize the average performance across 5×2 cross-validation trials for each dataset. Table 4 and 5 contain the results of conducting the Wilcoxon Signed-Rank Test on the results of all validation trials. This is a non-parametric statistical test designed to compare groups of paired data. In particular, it provides an alternative to the paired-t test when the data cannot be assumed to come from a normal distribution, as is the case.

For each dataset, we ran the Wilcoxon Signed-Rank test to determine whether there were statistically significant differences between the results of the Null model, the Unpruned Tree model and Pruned Tree model (pairwise). Our alternative hypothesis is that the Unpruned Tree model performs better than the Null model, and the Pruned Tree model performs the best out of all of them. The table contains the p-values for each test, and is color-coded to highlight the results that are statistically significant ($p < 0.05$).

Table 6 contains a comparison of the average number of nodes found in each tree for each of the validation trials, before and after pruning, as well as the average percent reduction in tree size due to pruning.

7.1 Hypothesis 1: Unpruned Decision Trees outperform Null Model

Across all datasets except *Forest Fires*, the Unpruned Decision Tree models consistently outperformed the Null model. This conclusion is supported by Wilcoxon signed-rank test,

which showed statistically significant differences ($p = 0.001$) between the performance of these models.

Generally speaking, the performance gap was substantial. For example, in *Breast Cancer*, the Unpruned Tree achieved an average accuracy of 0.938, compared to the accuracy of 0.655 from the Null model. In *Machine*, average MSE was drastically reduced from 25663 to 7143. Similar improvements were observed in *Car* and *Congress*, while *Abalone* showed a more modest but still significant improvement.

However, *Forest Fires* was a clear outlier. Here, the average MSE rose from 1.9518 under the Null Model to 3.7625 with the Unpruned tree model, a significant degradation in performance. This degradation is likely due to the extreme skew of the dataset, with most area values at or around 0, even after the log transformation, and several large outliers. In such cases, using information gain ratio as a splitting criterion may not be the optimal choice, and several alternative splitting criteria have been proposed. For instance, Hellinger distance has been shown to produce trees that are less skew-sensitive (Cieslak, Chawla, & Striegel, 2012). Exploring this approach in the feature could lead to improved performance on the Forest Fires dataset.

7.2 Hypothesis 2: Pruned Decision Trees outperform Unpruned Trees

For every dataset except *Car*, pruned trees achieved higher performance than unpruned trees. The improvements were generally less marked for the classification tasks. For instance, in *Breast Cancer*, pruning reduced the average error rate from 0.0618 to 0.0495, and in *Congress* from 0.0616 to 0.0506. Despite these seemingly modest gains, the Wilcoxon test confirms that they are statistically significant. Yet *Car* showed no meaningful change, with the pruned tree achieving nearly identical (but slightly worse) performance. It is interesting to note that out of the classification datasets, *Car* was the least affected by pruning, with its decision tree reduced by only 37%, compared to over 50% in the other ones. This suggests that there may be a link between pruning size reduction and performance increases, a relationship we explore in the third hypothesis.

In the regression datasets, the performance increases were more dramatic. For example, *Abalone* saw a reduction in average MSE from 9.1850 to 6.5788, and *Forest Fires* dropped from 3.7625 to 2.5902. These gains were again statistically significant for all regression datasets.

Overall, pruning proved to be an effective method for improving generalization and simplifying decision trees, although its benefits can clearly vary by dataset. In cases where the unpruned tree already generalizes well, or pruning doesn't effectively cut down the complexity of the tree, pruning may offer limited performance gains.

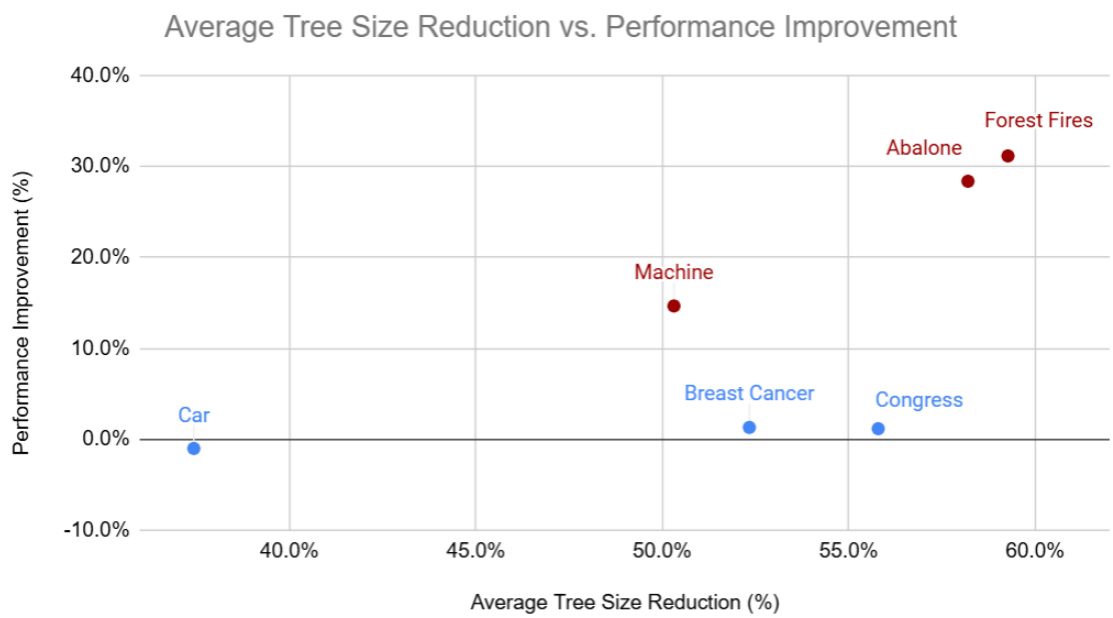
7.3 Hypothesis 3: Tree Size Reduction Correlates with Performance Improvement

In the introduction, we posited a positive correlation between the percentage reduction in tree size (as measured by number of nodes) and the relative performance improvement from pruning. Across the six datasets, we computed both the tree size reduction and the performance change from pruning, then plotted and analyzed the correlation.

Table 7: Pearson correlation between tree size reduction and performance improvement

Dataset Group	**Correlation Coefficient (r)**
Classification	0.973
Regression	0.999
Combined	0.658

Figure 1: Scatterplot of average tree size reduction vs. performance improvement from pruning: Classification datasets (blue) and regression datasets (red).



As shown in Figure 1 and Table 7, we observed a strong positive linear correlation within each task type:

- For regression datasets, the Pearson correlation coefficient was $r = 0.999$, indicating an almost perfect linear relationship.
- For classification datasets, the correlation was similarly strong at $r = 0.973$.

- However, when combining all six datasets, the overall correlation dropped to $r = 0.658$.

This discrepancy arose because classification and regression tasks displayed very different behaviors: while both showed positive slopes, the magnitude of performance improvement for a given amount of pruning varied significantly between the two. Regression datasets tended to see large performance gains with pruning, while classification datasets showed smaller gains despite similar reductions in tree size.

The scatterplot in Figure 1 confirms this: regression datasets cluster above the classification datasets, with both groups following their own linear trends. This suggests that that pruning may offer a clearer performance benefit for regression tasks than for classification tasks. However, we must stress that with only three data points per task type, the correlation estimates are highly sensitive to outliers. Further experiments on a wider variety of datasets would be needed to validate this trend.

8 Conclusion

In this project, we evaluated the performance of decision tree models across six diverse datasets, comparing Unpruned trees and Pruned trees to a Null Model. The results of our study confirmed that decision trees significantly outperform the Null Model in nearly all cases, with pruning often yielding further performance improvements while substantially reducing tree size. Although pruning did not universally improve performance, it consistently simplified models by reducing the number of nodes by by between 37% and 59%.

We also observed a strong positive correlation between tree size reduction and performance gain when classification and regression tasks were analyzed separately. However, when aggregated, this relationship weakened due to different scaling across task types. This suggests the value of pruning may be dependent on the task type, as well as the details of the datasets.

All in all, our results support pruning as a valuable technique for reducing overfitting and improving generalization in decision trees, although future work might explore alternative pruning criteria or skew-insensitive splitting metrics to further enhance robustness on challenging datasets.

References

[1] Cieslak, D. A., Chawla, N. V., & Striegel, A. (2012). Hellinger distance decision trees are robust and skew-insensitive. *Data Mining and Knowledge Discovery*, 24(1), 21–49. https://doi.org/10.1007/s10618-011-0212-1